

1. Title

Weather Explorer: A Dynamic 3D Urban Simulation with Real-Time Atmospheric and Air Quality Visualization.

2. Team Member List

Name	ID	Contribution
Md. Tasauf Islam	23-54025-3	Feature list, Conclusion
Mehereen Khan Mila	23-53560-3	Introduction, Description

3. Introduction

The rapid urbanization and industrial growth of the current century have brought environmental monitoring to the forefront of global concerns. One of the most pressing issues in modern metropolitan areas is the deteriorating quality of air, often quantified by the Air Quality Index (AQI). While data is frequently available in numerical formats, there is a significant gap in how the general public perceives the tangible impact of pollution on their immediate surroundings. Our project, "Weather Explorer: A Dynamic 3D Urban Simulation with Real-Time Atmospheric and Air Quality Visualization," aims to bridge this gap by providing an immersive, interactive platform that visualizes the interplay between urban infrastructure and atmospheric conditions.

A primary motivation for this project is the critical environmental state of cities like Dhaka, Bangladesh. Dhaka consistently ranks among the most polluted cities globally, frequently recording "Unhealthy" or "Hazardous" AQI levels. In such high-density urban environments, the presence of particulate matter creates thick, brownish smog that reduces visibility and alters the aesthetic and physiological experience of the city. By simulating these specific conditions, ranging from the clear blue skies of a low-AQI day to the dense, suffocating haze of a polluted afternoon, this project provides a visual narrative of the "silent killer" that is air pollution.

The scope of this project involves creating a procedural 3D city block where environmental parameters can be toggled in real-time. Similar recent projects in the field of computer graphics have focused on hyper-realistic weather rendering or static architectural visualizations. However, Weather Explorer introduces novelty by linking functional graphics techniques, such as particle systems for rain and snowfall and fixed-function fog for smog simulation, directly to environmental data points.

The overall simulation follows a day in the life of a digital city, where the user acts as an observer with the power to manipulate time and toxicity. Through the use of OpenGL and GLUT, the project demonstrates how mathematical transformations and rendering techniques can be applied to solve the real-life problem of making abstract environmental data visible and understandable to the human eye.

4. Description of the environment and list of methods

The development of Weather Explorer will be developed within a C++ programming environment by using the OpenGL for high-performance 3D rendering. Basic computer graphics techniques will be applied to model the solar system. To facilitate window management and user interaction, the GLUT (OpenGL Utility Toolkit) will be utilized. Development and testing will be performed in the CodeBlocks on a Windows operating system.

The simulation utilizes several fundamental computer graphics methods to achieve its dynamic visuals. Procedural modeling algorithms will be used to generate a 3D city grid, allowing for varied building heights and positions without manual vertex entry. The implementation relies heavily on coordinate transformations, using functions to position urban elements and animate moving objects like vehicles. By dynamically mapping color and density parameters to pollution levels, the system provides a clear visual representation of environmental health, ranging from clear blue skies to hazardous brownish-yellow smog.

Furthermore, weather phenomena such as rain and snow will be managed through an array-based particle system that updates vertex positions in real-time to simulate gravity and wind vectors. Lighting and material properties are also utilized to simulate the transition between day and night, affecting how building surfaces reflect ambient and diffuse light. To ensure smooth, flicker-free animations during these real-time transitions, double buffering is employed. Overall, the project integrates these methods to address the real-life problem of environmental visualization, providing an interactive scope for users to explore the relationship between urban density and atmospheric conditions.

5. Feature set

The simulation generates a procedural city grid where various environmental factors specifically weather conditions and air quality can be manipulated in real-time to observe their visual impact on a metropolitan area. By integrating coordinate transformations for object movement, fixed-function fog for smog visualization, and particle systems for atmospheric effects, the project transforms abstract environmental data into a tangible visual experience. The following features will be implemented using CodeBlocks on a Windows operating system.

- 1) **Road Median:** A thin yellow strip used to divide traffic lanes.
- 2) **Sidewalk Quads:** Raised grey planes rendered on both sides of the road.
- 3) **Green Zone Planes:** Large green rectangular quads representing park areas.
- 4) **Varied Skyscrapers:** Buildings with procedural heights generated through loops.
- 5) **Apartment Blocks:** Wider, shorter cuboids representing residential housing.
- 6) **Streetlight Post:** A vertical thin cuboid used for city lighting infrastructure.
- 7) **Streetlight Head:** A horizontal small block attached to the top of each pole.
- 8) **Central AQI Monument:** A unique 3D pillar structure located at the city center.
- 9) **Building Windows:** Small, contrasting colored quads placed on building faces.

- 10) **Boundary Walls:** Thin vertical planes defining the perimeter of the park.
- 11) **Simple Bench Base:** Two small blocks rendered to serve as bench legs.
- 12) **Bench Seating:** A flat horizontal block placed atop the bench legs.
- 13) **Water Body Plane:** A blue rectangular plane representing an urban lake.
- 14) **Rain Particle Generation:** Creating an array of line primitives at randomized top-screen coordinates.
- 15) **Rain Gravity Simulation:** Programming the downward vertical movement of rain lines.
- 16) **Rain Recycling:** Resetting rain particles to the top of the screen upon ground impact.
Fog Activation: Enabling the glFog function to simulate atmospheric haze.
- 17) **AQI Density Scaling:** Increasing fog density based on user input to simulate heavy smog.
- 18) **AQI Color Shifting:** Changing fog color from grey to brown to represent hazardous air.
- 19) **Lightning Flash:** Temporarily changing the background glClearColor to white.
- 20) **Daytime Skybox:** Setting the background color to light blue.
- 21) **Nighttime Skybox:** Setting the background color to dark navy.
- 22) **Sunset Skybox:** Setting the background color to an orange and red gradient.
- 23) **Vehicle Base:** Drawing a rectangular block representing a standard car.
- 24) **Vehicle Movement:** Translating the car block along the road axis.
- 25) **Vehicle Loop:** Resetting the car to the start of the road after it exits the field of view.
- 26) **Nighttime Headlights:** Drawing two small yellow spheres on the front of vehicles.
- 27) **Headlight Toggle:** Turning car lights on or off based on the simulated time of day.
- 28) **AQI Monument Scaling:** Using glScale to change the monument height based on pollution levels.
- 29) **Window Illumination:** Changing window quad colors to yellow during night mode.
- 30) **Camera Rotation (Left):** Rotating the entire scene via the left arrow key.
- 31) **Camera Rotation (Right):** Rotating the entire scene via the right arrow key.
- 32) **AQI Increase Key:** Programming the 'f' key to raise the pollution levels.
- 33) **AQI Decrease Key:** Programming the 'F' key to lower the pollution levels

6. Conclusion

The project provides a functional 3D simulation that effectively visualizes the relationship between urban environments and atmospheric health. By utilizing the OpenGL and GLUT framework, the project successfully translates abstract environmental data, such as the Air Quality Index (AQI), into a tangible visual experience through dynamic fog and particle systems.

While the current version addresses the real-life problem of environmental awareness, it has certain limitations, such as simplified geometric models for buildings and basic vehicle logic. Future development could involve integrating real-time weather APIs to reflect actual conditions in cities like Dhaka and adding more advanced lighting effects. Ultimately, this simulation demonstrates the power of computer graphics in creating educational tools that make complex environmental issues visible and understandable.